Development Engineering Project
CP-301

Mid-Sem Report

Prediction of Faults in Machines based on ML

Department of Mechanical Engineering, IIT Ropar

Supervised by

Prof. Navin Kumar

Submitted by:

Sahil Mohal – 2022MEB1343
Shreyas Manak -2022MEB1345
Tarini Magotra -2022MEB1358
Vedant Gupta -2022MEB1364

Submitted on

05-03-2024

1) Introduction

Bearings are critical components in countless mechanical systems, from industrial machinery and automobiles to aerospace and energy sectors. They support rotating shafts, reduce friction, and ensure smooth operation. If a fault occurs in the system, it can severely affect the process, increasing its bottleneck affecting productivity and safety.
Vibrations play a crucial role in analyzing the health and performance of rotating machinery, Rotating components can increase vibrations in a system. While some level of vibration is inevitable and normal, excessive or irregular vibrations often indicate underlying mechanical issues, such as misalignment or imbalance occurring due to spalling, pitting or minute cracks. Understanding these vibrations and delving into its causes can help increase the efficiency of our system. Stator current responses can directly indicate faults in a mechanical system. One of the most common types of faults in the system arises from bearing failures. Predicting these faults can have a significant impact on system reliability, efficiency, and maintenance costs. Early detection of bearing faults allows for early maintenance ensuring safety.

2) Problem Description

Problem Statement - Prediction of faults in machines based on ML

Detecting and diagnosing these faults early is crucial for ensuring the reliability, efficiency, and longevity of industrial systems.
In this project, we aim to detect faults in mechanical systems with major focus on bearing faults especially those occurring in outer rings. The faults in the bearings were artificially introduced by using EDM(Electronic Discharge Machine) in the form of holes of varying diameters. Our methodology involves conducting detailed vibration analysis, applying MCSA and FFT techniques, and utilizing machine learning (ML) models to predict faults based on normal and faulty system characteristics.
Through this approach, we aim to develop an intelligent fault detection system that can effectively classify and predict mechanical failures, enabling proactive maintenance and minimizing system failures.

Objectives:

- Analyze vibrations and motor current signals to study system behavior under healthy and faulty conditions.
- Introduce controlled bearing faults (e.g., holes of different diameters) and examine their impact on system performance.
- Perform Vibration Analysis to detect variations in system response caused by bearing defects.
- Utilize Motor Current Signature Analysis (MCSA) to observe electrical signal variations linked to mechanical faults.
- Apply Fast Fourier Transform (FFT) to extract frequency-domain characteristics for fault identification.
- Train Machine Learning (ML) models using normal and faulty condition data to predict and classify faults.
- Compare different ML algorithms to determine the most effective model for fault prediction.
- Develop a predictive maintenance framework to prevent unexpected failures and optimize system performance.

3) Existing Studies and Literature Review

Detection of Bearing Faults in Mechanical Systems:

Detecting faults in mechanical systems, particularly in bearings coupled with shafts, plays a pivotal role in reducing downtime and improving operational efficiency in industrial applications. Various methods have been explored, ranging from traditional vibration-based analysis to more modern approaches utilizing motor current signature analysis (MCSA). Advancements in signal processing techniques, coupled with machine learning algorithms, have shown promise in improving fault detection accuracy and reliability.

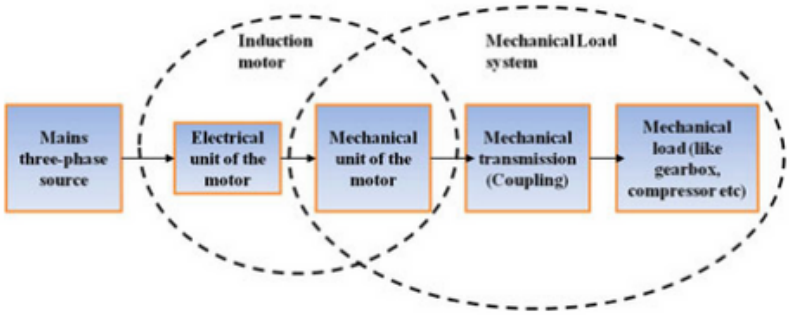


Fig 1: Constituent parts of a mechanical system coupled with an induction motor.

1. Utilizing Vibration-Based Techniques:

The research highlights the effectiveness of vibration-based methods for detecting bearing faults. Vibration analysis provides direct insights into how bearing faults generate characteristic vibration signatures, which can be captured and analyzed to detect anomalies. For our project, this reinforces the importance of incorporating vibration sensors in our system to measure vibrations and perform fault diagnosis. It also gives us a baseline for understanding the vibration signatures of healthy versus faulty bearings, which will be crucial in distinguishing between normal and abnormal conditions in our system.

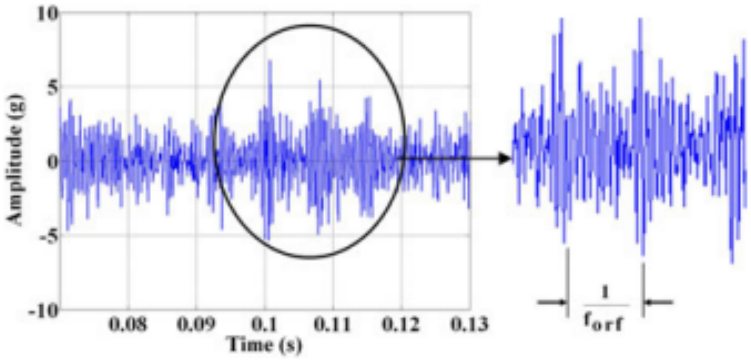


Fig 2: Vibration time domain signature of an ORF bearing.

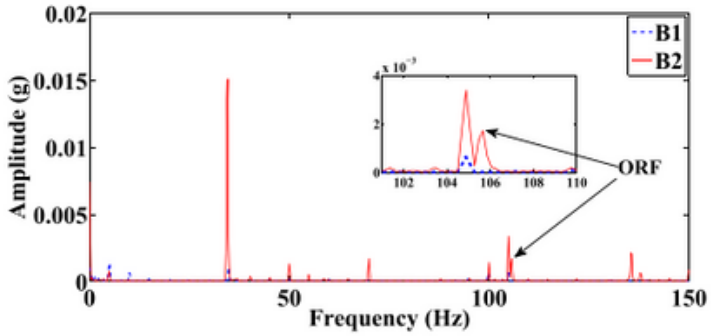


Fig 3: Frequency spectrum (vibration signature) of a healthy and faulty bearing.

2. Adapting Motor Current Signature Analysis (MCSA):

While MCSA is traditionally used for motor fault detection, its application to bearings coupled with shafts, as explored in the research, is highly relevant to our project. The research demonstrates that faults in bearings can affect the torque and the current signature of the motor, leading to characteristic modulations in the stator current. This insight allows us to consider MCSA as a potential diagnostic tool for detecting faults in bearings coupled to shafts, without the need for complex and expensive vibration sensors. MCSA offers a non-intrusive, cost-effective solution, especially beneficial in systems where vibration sensors might be difficult to install or maintain.

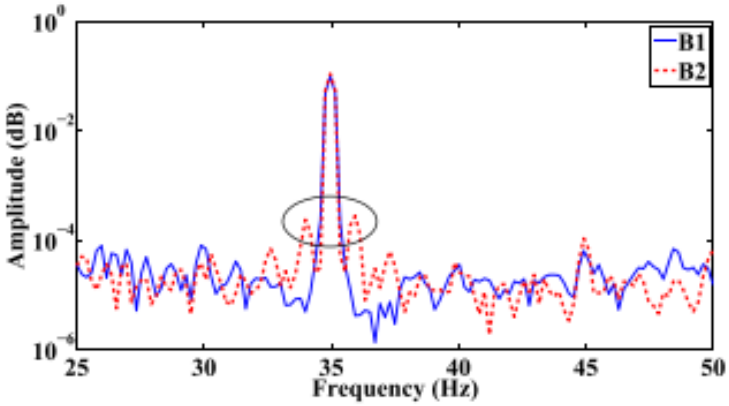


Fig 4: Comparison of current spectra of a healthy and faulty bearings at 35 Hz. B1 and B2 refer to healthy and faulty bearing data respectively.

Additionally, the research provides valuable information on how fault-related frequencies appear as sidebands in the frequency spectrum of the stator current. This can be used in our project to design algorithms that analyze the stator current and detect fault frequencies, improving our ability to diagnose faults in bearings without requiring additional hardware or costly sensors.

3. Machine Learning-Based Enhancement:

Integrating machine learning with MCSA and vibration analysis enhances fault detection accuracy in our project. By training models like SVMs, Random Forests, and ANNs on features extracted from vibration and stator current signals, we can classify different bearing faults, improving system precision and adaptability to varying conditions. This approach reduces manual inspection and boosts maintenance efficiency.

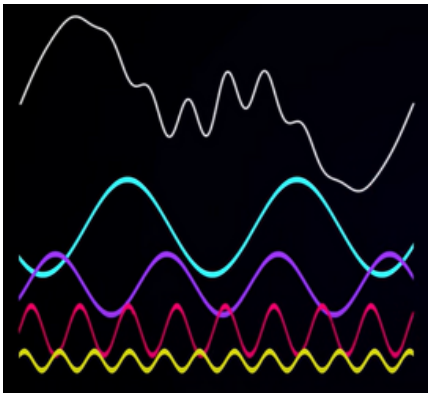
4. Hybrid Approaches for Robust Fault Detection:

This paper focuses on using MCSA and vibration analysis to detect bearing faults, which helps us understand how these methods can work together, especially in noisy environments. It shows how we can use motor current and vibration data to identify faults without needing complex sensors.

In our project, we're building on this by adding machine learning. While the paper helps us understand MCSA and vibration analysis, we plan to use machine learning models like SVMs and ANNs to make our system smarter, more accurate, and capable of automatically detecting and classifying faults under different conditions.

5. Explanation of FFT Analysis

Time Frequency Duality:



A fast Fourier transform (FFT) is an algorithm that computes the discrete Fourier transform (DFT) of a sequence, or its inverse (IDFT). A Fourier transform converts a signal from its original domain (often time or space) to a representation in the frequency domain and vice versa. FFT suggests if we are capable of Decomposing a wave into a combination of sine and cosine waves of variable frequencies and amplitudes and we can get corresponding contribution of all frequencies converting the graph from time domain to frequency domain.

Heisenberg principle suggest that $\triangle t . \triangle f > 1$ hence making it impossible to obtain a good resolution of both and hence we only obtain it in frequency domain

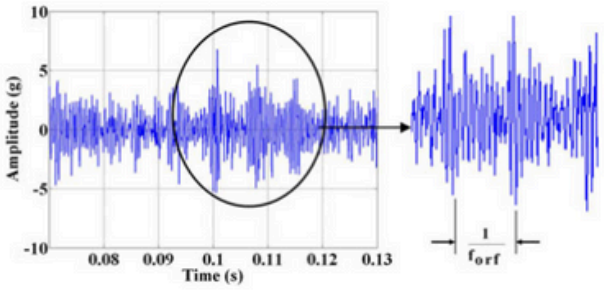


Fig 5: Vibration time domain signature of ORF bearing.

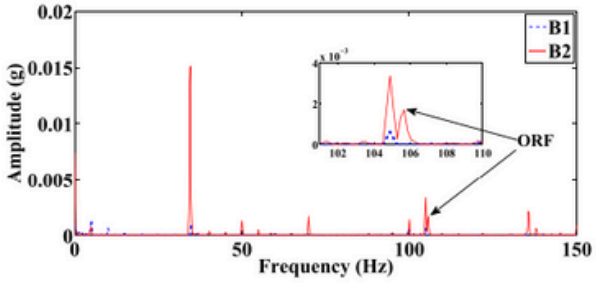


Fig 6: Frequency spectrum (vibration signature) of a healthy and a faulty bearing.

Vibration Time Domain Signature of an ORF Bearing

The plot shows amplitude (g) versus time (s), representing the vibration response of a bearing with an outer race fault. The recurring fluctuations in amplitude detect the presence of an ORF defect and its periodic impacts.

Frequency Spectrum (Vibration Signature) of a Healthy and Faulty Bearing

The plot shows amplitude (g) versus frequency (Hz), illustrating the frequency components of vibrations in both a healthy and faulty bearing. The peaks in the spectrum indicate dominant vibration frequencies, with additional peaks at characteristic fault frequencies associated with the outer race fault (ORF).

The two curves (B1 and B2) correspond to different conditions (healthy vs. faulty), showing how fault-induced vibrations introduce additional frequency components.

4) Experimental Setup and Description

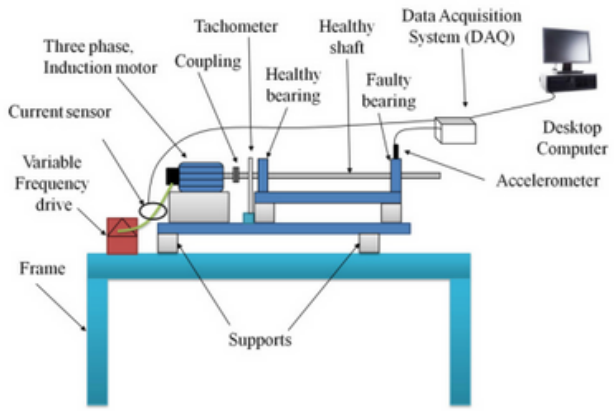


Fig 7: Diagram of the setup

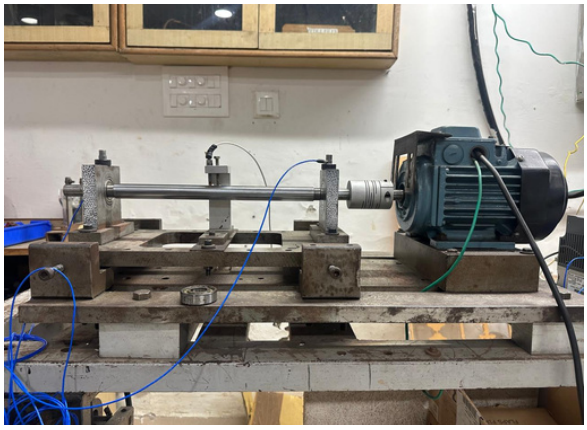


Fig 8: Original Apparatus

- The setup consists of a two-pole, three-phase, 1 H.P. (0.75 kW) induction motor.
- The motor is coupled to a mild steel shaft of 25 mm diameter, which is mounted on two identical deep groove ball bearings fixed in two plummer blocks.
- A torsionally stiff spring coupling connects the induction motor to the shaft.
- A Variable Frequency Drive (VFD) is attached to the motor to adjust its speed. The frequency spectrum suitable for our project varied from 20Hz to 40Hz.
- The motor bearings are healthy as of the original setup.
- A pair of reverse dial gauges is used to align the rotor shaft with the motor shaft within 10 microns.
- For fault detection using Motor Current Signature Analysis (MCSA), faulty bearings are installed in the mechanical system.
- The bearing faults were artificially created using Electrical Discharge Machining (EDM).
- The ball bearings used in the study are 6204-ZZ series, make SKF .

a) Bearing Healthy vs Faulty



Fig 9: Faulty vs Healthy Bearing

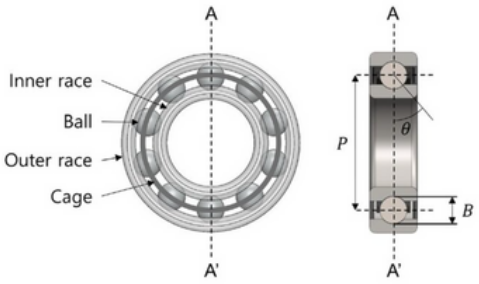


Fig 10: Diagram of a Healthy Bearing

Faults introduced on outer ring in the form of holes of diameters of 2mm and 4mm to detect faults.

Bearing Components

- Outer Ring – The stationary part of the bearing that fits into the housing and provides raceway support.
- Inner Ring – The rotating part that mounts onto the shaft and holds the rolling elements in place.
- Rolling Elements – Balls or rollers that reduce friction by rolling between the inner and outer rings.
- Cage – A separator that evenly spaces the rolling elements, preventing contact and ensuring smooth rotation.

b) 2 Pole 3 Phase Induction Motor Power Rating 1 HP

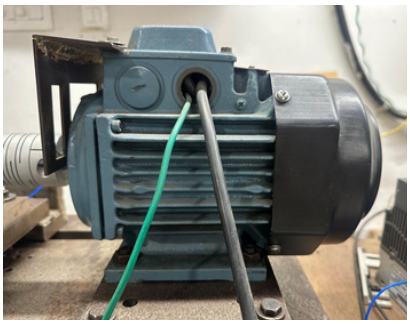


Fig 11: 2 Pole 3 Phase Induction Motor Power Rating 1 HP

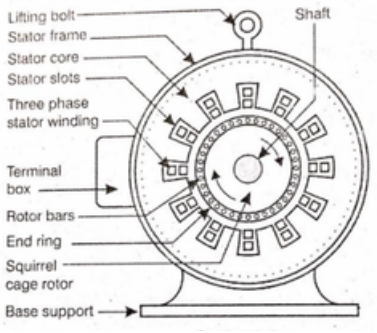


Fig 12: Sectional view Line Diagram of 3 phase Motor

We used a two-pole three-phase induction motor with a power rating of 1 HP (0.75 kW) for the experiments. This motor is coupled to a mild steel shaft mounted on two identical deep groove ball bearings (SKF 6204-ZZ series) fixed in plummer blocks. The bearings used for fault detection are installed in the mechanical system, while the induction motor bearings are assumed to be healthy.

Healthy vs Faulty Bearings: The amplitude of the fundamental running frequency (35 Hz) and its harmonics are higher for faulty bearings compared to healthy ones. This indicates that bearing faults introduce anomalies in the current signal due to rotor displacement relative to the stator.

Characteristic Fault Frequencies: Faults produce impulses at characteristic frequencies related to bearing geometry and rotor speed. These frequencies manifest as sidebands around the running frequency in the current spectrum, which are reliable indicators of bearing defects.

The induction motor acts as a transducer, reflecting load variations caused by bearing faults in its stator current spectrum. By monitoring current harmonics and analyzing sidebands using techniques like FFT and wavelet scalograms, the study effectively distinguishes between healthy and faulty bearings installed in mechanical systems driven by an induction motor.

c) Coupling

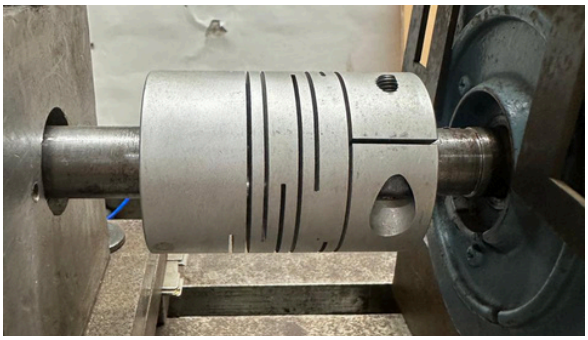


Fig 13: Coupler

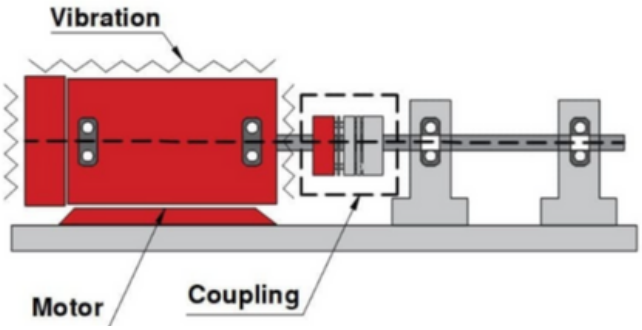


Fig 14: Diagram of a coupler

Couplers are widely used in mechanical systems to connect two rotating shafts while allowing for:

- **Torque Transmission:** They enable power transfer between a driving and driven component (e.g., motor and pump or motor and gearbox).
- **Alignment Adjustment:** Flexible couplings can compensate for angular or axial misalignments between shafts, reducing wear and tear on connected components.
- **Shock Absorption:** Couplers can dampen vibrations and absorb shocks to protect sensitive equipment.
- **Load Transfer:** It allowed uniform transfer of mechanical load from the motor to the bearings, ensuring consistent conditions for analyzing healthy and faulty bearing performance.

d) DAQ-Data Acquisition System



Fig 15: DAQ



Fig 16: DAQ ports

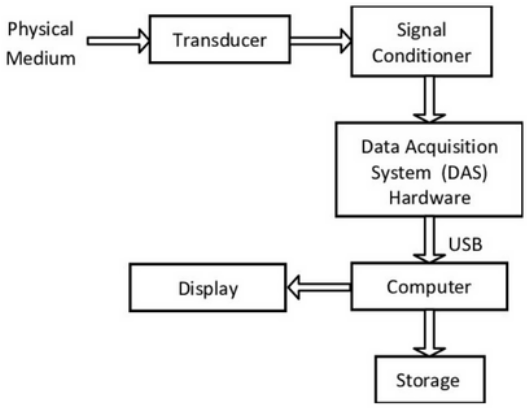


Fig 17: Block Diagram of DAQ

DAQ stands for Data Acquisition, which refers to the process of collecting, measuring, and analyzing data from physical phenomena or systems. A DAQ system typically consists of sensors, signal conditioning hardware, analog-to-digital converters (ADC), and software to process the data. It is widely used to monitor and control various parameters such as temperature, pressure, voltage, current, and more in real-time.

Components of a DAQ System:

- Sensors: Convert physical phenomena (e.g., temperature, pressure) into electrical signals.
- Signal Conditioning: Prepares the sensor signals by amplifying, filtering, or isolating them.
- Analog-to-Digital Converter (ADC): Converts analog signals into digital data for processing.
- DAQ Hardware: Interfaces with sensors and ADCs to collect data.
- Software: Used for data visualization, analysis, and storage.

e) Probe



Fig 18: Hall Effect Probe

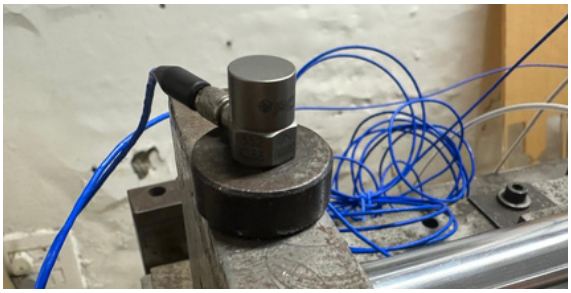


Fig 19: Accelerometer Probe

A probe is a device or instrument used to sense, measure, or test physical quantities such as electrical signals, temperature, pressure, or vibrations. Probes are designed to interact with the system being tested without significantly altering its behavior. They act as intermediaries between the system and the data acquisition or monitoring equipment.

- Hall Effect Current Probes: These were used to measure the input current waveform of the induction motor's three phases. They provided non-invasive current measurement by detecting the magnetic field generated by the current flow.

- Accelerometer Probe: Used to measure vibrations on the plummer block where the bearings were mounted. This data was used to validate findings from Motor Current Signature Analysis (MCSA).

f) Shaft

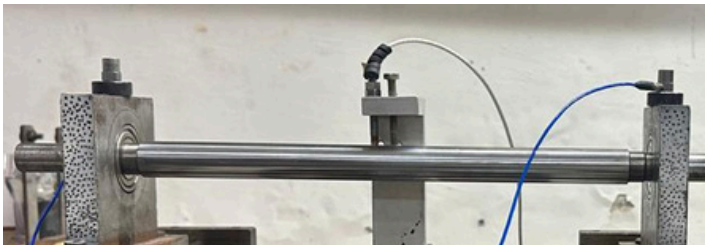


Fig 20: Shaft

A shaft is a mechanical component that is typically cylindrical and used to transmit rotational motion and torque from one part of a machine to another. It serves as a fundamental element in mechanical systems, enabling the transfer of power between components such as motors, gears, pulleys, and bearings.
Shafts are usually made from materials like steel, stainless steel, or alloys to provide the strength and durability needed to withstand high rotational speeds, loads, and stresses.

5) Methodology

Step 1: Experimental Setup

1. Induction Motor: A two-pole, three-phase induction motor (1 HP) drives the mechanical system.
2. Mechanical System: A mild steel shaft mounted on two identical deep groove ball bearings (SKF 6204-ZZ series) housed in plummer blocks.
3. Coupler: A flexible coupling connects the motor shaft to the driven shaft.
4. Sensors:
   Hall Effect current probes measure stator current signals from the motor's three phases.
   An accelerometer measures vibrations on the plummer block.

Step 2: Data Acquisition

1. A National Instruments (NI) DAQ system records signals:
2. Stator current signals are sampled at 5.2k for a duration of 5.12 seconds (26626 samples per experiment).
3. Vibration data is collected simultaneously for comparison

The raw data is stored for further processing.

Step 3: Signal Processing

1)FFT Analysis:

- The Fast Fourier Transform (FFT) is applied to the stator current signals to identify frequency components.
- Harmonic amplitudes and sidebands around the running frequency (35 Hz) are analyzed for fault detection.

Step 4: Feature Extraction

Using signal processing technique (FFT), we extracted meaningful features from stator current signals:

1) Frequency domain features for current signature:

- Peak current frequency: It is the frequency at which the magnitude of the current signal's FFT is at its absolute maximum for a given window of data.
- BPFO (Ball Pass Frequency Outer race): This is the fundamental frequency generated by a defect on the outer race of a bearing. As rolling elements pass over the defect, they create impacts at this characteristic frequency.
- Modulation: When a bearing defect frequency (like BPFO) is present, it can be "modulated" by another frequency, typically the rotational speed of the shaft. This means the amplitude of the BPFO signal varies at the rate of the shaft's rotation. This amplitude modulation creates new frequencies in the spectrum, which are the sidebands.
- Upper Sideband BPFO: Upper Sideband BPFO: This refers to a frequency that is equal to the BPFO plus the modulating frequency.
- Lower Sideband BPFO: This refers to a frequency that is equal to the BPFO minus the modulating frequency.

Step 5: Data Preparation

The process begins by taking raw time-domain data for vibration and, if available, current signals. This data is segmented into overlapping windows. A Hanning window function is applied to each segment to minimize spectral leakage before performing a Fast Fourier Transform (FFT) to convert the signals to the frequency domain. The features are then extracted from the magnitude spectrum of these FFTs. Features derived from the vibration signal are prefixed with 'fft_' , and those from the current signal are prefixed with 'cur_'.

III. Detailed Data Preparation Steps

A. Window Creation and Metadata Association

1. Window Segmentation Strategy:
    a. The `step` size for moving from one window to the next is calculated based on `window_size` and `overlap`: `step = int(window_size * (1 - overlap))`.
    b. The raw data is first grouped by `op_freq_col`, `bpfo_col`, and `label_col`. This ensures that windows are created consistently within data segments that share the same operational parameters and fault status, preventing windows from spanning across different conditions.
2. Iterative Windowing:
    a. For each group of data:
        i. The vibration signal (`vib_signal`) and current signal (`current_signal`, if available) are extracted.
        ii. The code iterates through the signals, creating windows of `window_size` samples.
        iii. For each window, both the vibration segment (`vib_window`) and the corresponding current segment (`cur_window`) are stored.
3. Metadata Storage:
    a. Alongside each windowed signal pair, the following metadata is stored:
        i. A unique `window_id`.
        ii. The `op_freq` (operating frequency) for that window.
        iii. The `bpfo` value.
        iv. The `label` (fault status).

B. FFT Computation and Feature Extraction per Window

This stage iterates through each created window and its associated metadata.

1. Preprocessing - Hanning Window Application:
    a. To minimize spectral leakage (artifacts in the frequency spectrum due to abrupt start/end of finite signals), a Hanning window function (`np.hanning`) is applied to both the vibration (`vib_window`) and current (`cur_window`, if present) segments before FFT computation.

2. FFT Computation:
    a. The Real Fast Fourier Transform (`np.fft.rfft`) is computed for the windowed vibration signal (`windowed_vib`). This is suitable for real-valued input signals and returns only the positive frequency components.
    b. The magnitude of the FFT (`vib_fft_mag = np.abs(vib_fft)`) is calculated, as this represents the amplitude of each frequency component.
    c. The corresponding frequency bins (`freqs`) are calculated using `np.fft.rfftfreq`, which takes the window size and sampling interval (`1/sampling_rate`) as input.
    d. The same FFT computation process is applied to the windowed current signal (`windowed_cur`) if it's available, yielding `cur_fft`, `cur_fft_mag`.
3. Feature Extraction (using `extract_fft_features` helper function):
    a. The `extract_fft_features` function is called to calculate a set of features from the `vib_fft_mag` and `freqs`.
    b. If current signal data is present, `extract_fft_features` is called again for `cur_fft_mag` and `freqs`, with a prefix 'cur_' to distinguish these features.
    c. Cross-Signal Feature: If both vibration and current signals are available, a `vib_cur_coherence` feature is calculated. This feature measures the linear relationship between the two signals in the frequency domain: where is the FFT of vibration, is the FFT of current, and is the complex conjugate of .
4. Metadata Integration:
    a. The `window_id`, `op_freq`, `BPFO`, and `label` are added to the dictionary of extracted features for the current window.
5. Collection:
    a. The dictionary containing all features and metadata for the window is appended to a list `fft_features`.

Step 6: Machine Learning Model Development

1) The following machine learning algorithms were selected:
- Decision Tree
- Support Vector Machines (SVM)
- Artificial Neural Networks

2) Train the model:
- Used 60% of the labeled dataset for training.

3) Test and validate:
- Tested the model on 20% test dataset.
- Evaluated model performance on the 20% remaining unseen data using metrics like accuracy, precision, recall and F1-score.

- Accuracy: The ratio of correctly predicted observations to the total observations.
- Precision: The ratio of correctly predicted positive observations to the total predicted positives.
- Recall: The ratio of correctly predicted positive observations to all actual positives.
- F1 Score: The harmonic mean of precision and recall, providing a balance between them.

6) Results

I) Vibration analysis vs Current signature obtained at 30Hz

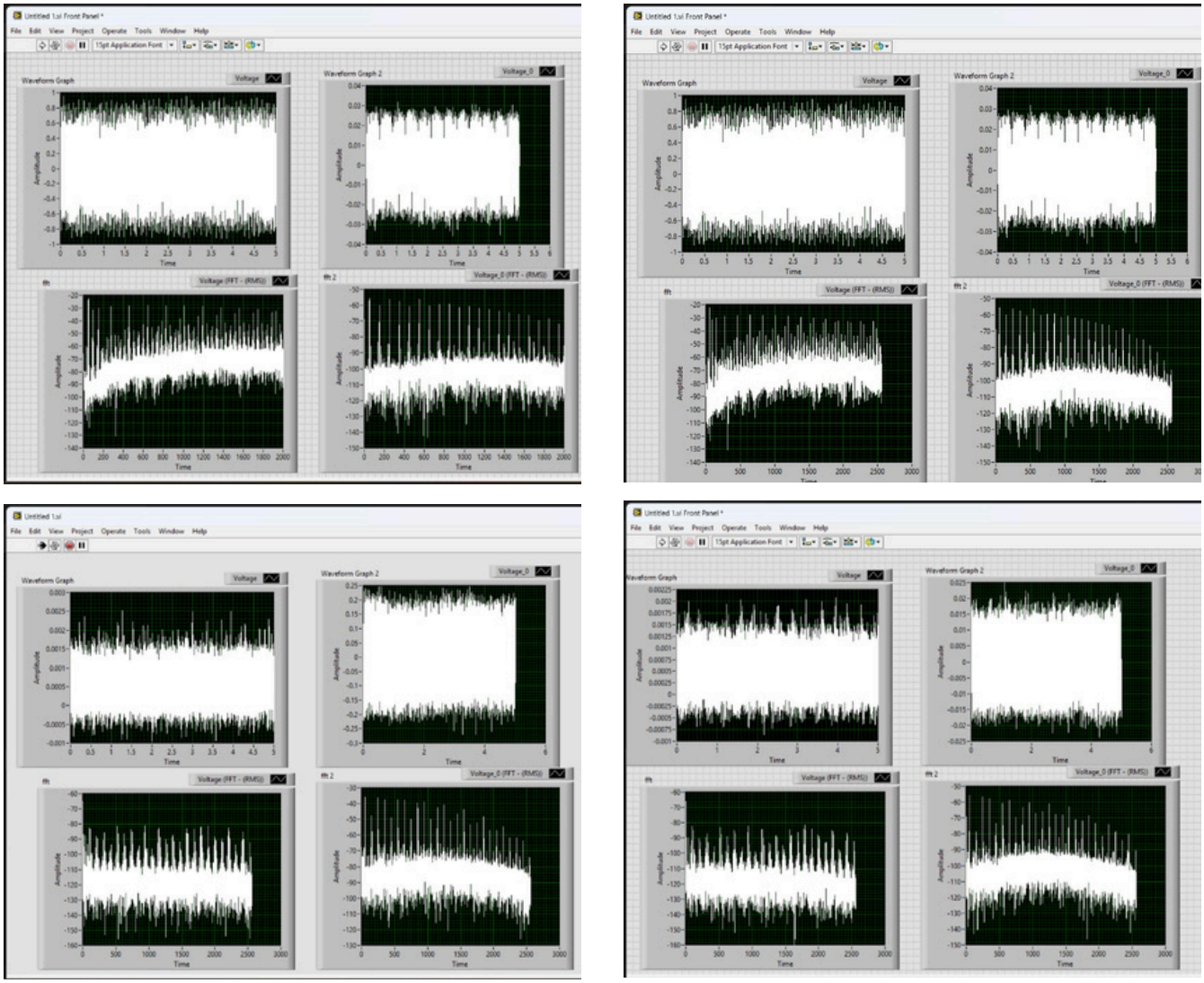


Fig 21: Data from 4 Experiments

The data presented above was acquired using a Hall effect current probe and an accelerometer mounted on the plummer block housing the bearing, capturing both time-domain and frequency-domain (FFT) signals. The collected data will serve as the foundation for classifying faults as part of our machine learning efforts.

II) Decision Tree: A decision tree has the ability to simplify complex decisions through a structured, step-by-step process. It splits data based on key features, forming branches that lead to outcomes. This approach enhances interpretability, making it ideal for classification or regression tasks in a clear, rule-based manner.

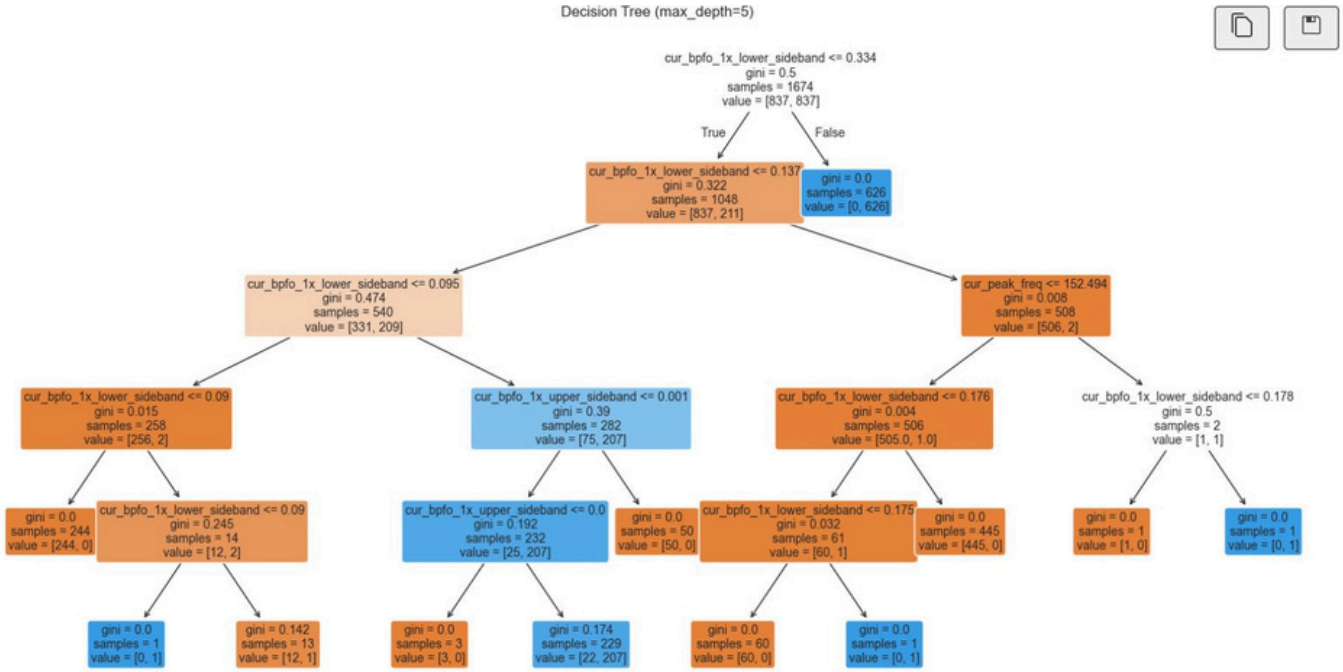


Fig 22: Flowchart for our decision tree model.

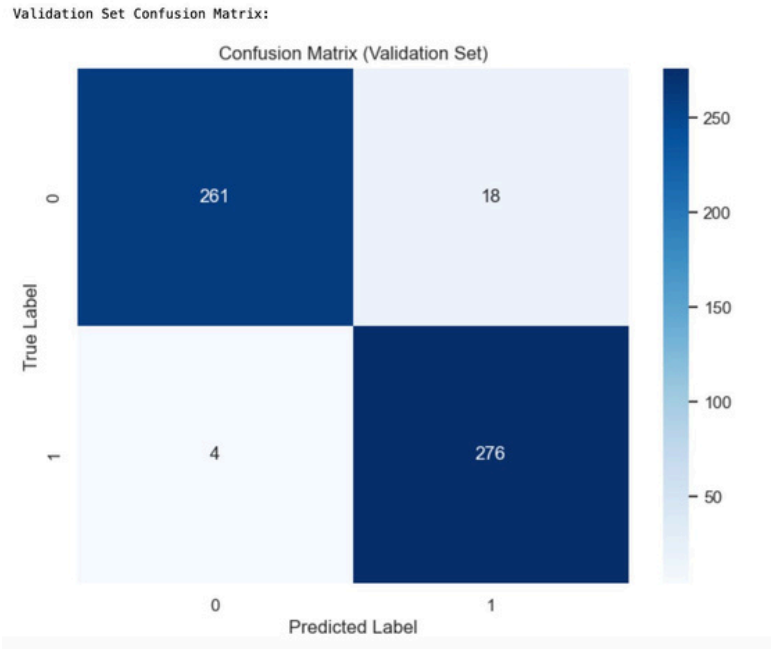


Fig 23: Confusion Matrix for validation dataset.

```
Training model with 3 selected features:
  1. cur_peak_freq
  2. cur_bpfo_1x_upper_sideband
  3. cur_bpfo_1x_lower_sideband

Model Training Complete:
  Validation Accuracy: 0.9606
  Test Accuracy: 0.9571

Validation Set Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.94      0.96       279
           1       0.94      0.99      0.96       280

    accuracy                           0.96       559
   macro avg       0.96      0.96      0.96       559
weighted avg       0.96      0.96      0.96       559
```

Fig 24: Final Performance Metrics (Decision Tree).

III) Artificial Neural Network: An Artificial Neural Network (ANN) is a machine learning model inspired by the human brain. It consists of layers of interconnected nodes that process data, learn patterns, and make predictions. ANNs are trained using algorithms like backpropagation and are effective for tasks such as classification, regression, and pattern recognition.

```
Training neural network with 3 selected features:
  1. cur_peak_freq
  2. cur_bpfo_1x_upper_sideband
  3. cur_bpfo_1x_lower_sideband

Model Structure:
NeuralNetwork(
  (model): Sequential(
    (0): Linear(in_features=3, out_features=128, bias=True)
    (1): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU()
    (3): Dropout(p=0.2, inplace=False)
    (4): Linear(in_features=128, out_features=128, bias=True)
    (5): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (6): ReLU()
    (7): Dropout(p=0.2, inplace=False)
    (8): Linear(in_features=128, out_features=128, bias=True)
    (9): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (10): ReLU()
    (11): Dropout(p=0.2, inplace=False)
    (12): Linear(in_features=128, out_features=2, bias=True)
  )
)
```

Fig 25: Structure of ANN used.

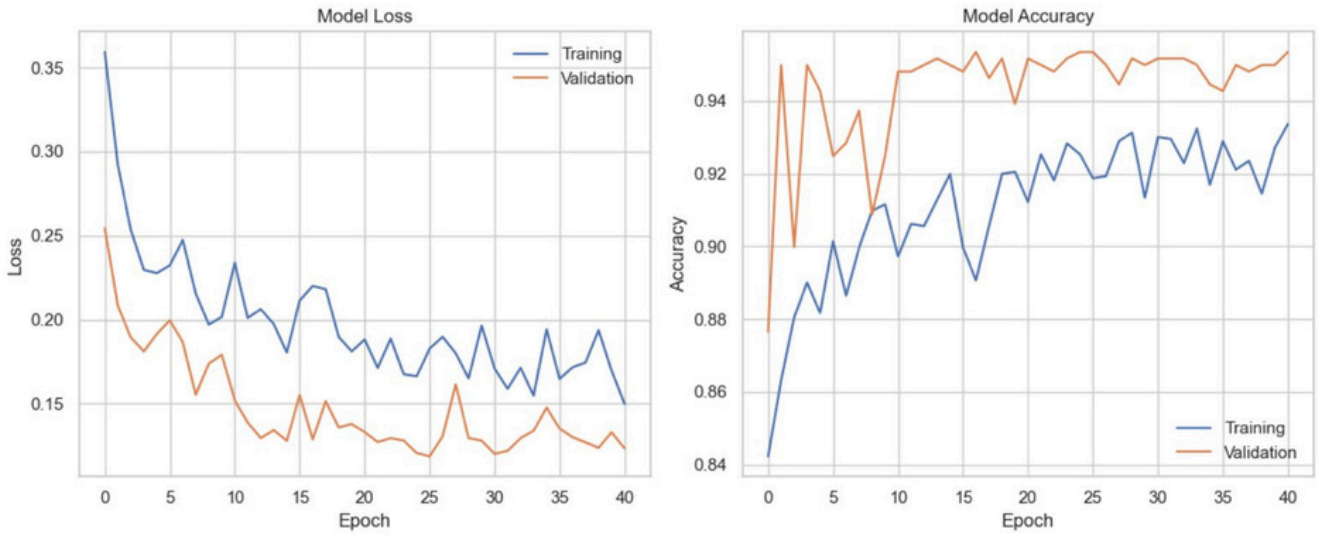


Fig 26: Loss and Accuracy for ANN model vs Epoch (iterations).

```
Neural Network Evaluation on Test Set:
Test Accuracy: 0.9445

Test Set Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.90      0.94       280
           1       0.91      0.99      0.95       279

    accuracy                           0.94       559
   macro avg       0.95      0.94      0.94       559
weighted avg       0.95      0.94      0.94       559
```

Fig 27: Performance Metrics (ANN).

IV) Support Vector Machine: A Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that best separates data into classes. SVM is effective in high-dimensional spaces and handles both linear and non-linear data using kernel functions for improved accuracy.

Validation Set Confusion Matrix:

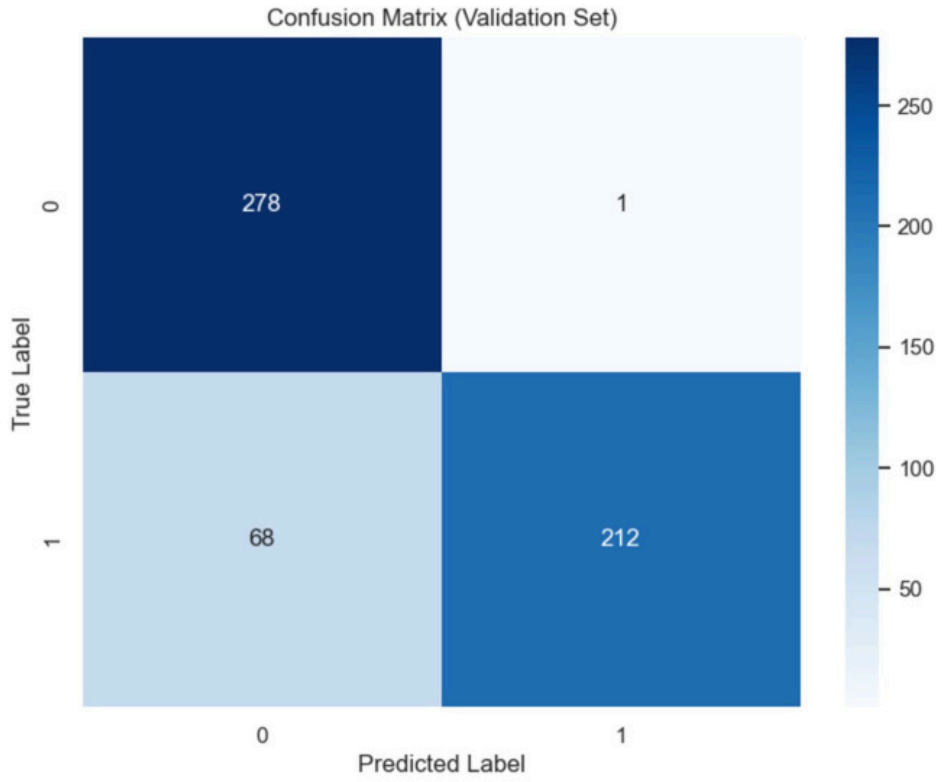


Fig 28: Confusion Matrix for validation dataset.

```
Training SVM model with 3 selected features:
  1. cur_peak_freq
  2. cur_bpfo_1x_upper_sideband
  3. cur_bpfo_1x_lower_sideband

SVM Parameters:
  Kernel: rbf
  C (Regularization): 1.0
  Gamma: scale

Model Training Complete:
  Validation Accuracy: 0.8766
  Test Accuracy: 0.8712

Validation Set Classification Report:
              precision    recall  f1-score   support

           0       0.80      1.00      0.89       279
           1       1.00      0.76      0.86       280

    accuracy                           0.88       559
   macro avg       0.90      0.88      0.87       559
weighted avg       0.90      0.88      0.87       559
```

Fig 29: Parameters and Performance Metrics (SVM).

7) Future Work:

In this study, we have collected vibration and motor current signature analysis (MCSA) data to detect bearing faults in mechanical systems. The analysis using Fast Fourier Transform (FFT) and signal processing techniques has provided critical insights into the frequency-domain characteristics of healthy and faulty bearings. To further enhance the fault detection capabilities, machine learning (ML) algorithms such as Decision Trees, Artificial Neural Networks (ANN), and Support Vector Machines (SVM) have been applied and evaluated for classification accuracy.

Building on this foundation, future work will focus on advancing towards real-time fault detection and predictive maintenance using the following approach:

1. Feature Extraction and Selection:
- Extract and refine time-domain and frequency-domain features from sensor data.
- Apply statistical, spectral, and wavelet-based methods to enhance feature relevance and reduce dimensionality.
2. Model Enhancement and Optimization:
- Further optimize the existing Decision Tree, ANN, and SVM models using hyperparameter tuning and ensemble strategies.
- Explore advanced deep learning models such as Convolutional Neural Networks (CNN) for improved feature learning.
3. Model Performance Evaluation:
- Continue evaluating model performance using metrics like accuracy, precision, recall, and F1-score.
- Use k-fold cross-validation and confusion matrix analysis to ensure reliability and robustness.
4. Deployment for Real-Time Predictive Maintenance
- Integrate the trained models into a real-time monitoring system capable of continuously analyzing vibration and MCSA signals.
- Develop a predictive maintenance framework that leverages ML predictions to forecast potential failures, schedule timely interventions, and minimize unplanned downtimes.
- This system will facilitate condition-based maintenance, reduce manual inspections, and extend machinery lifespan through data-driven insights.

By embedding machine learning into a predictive maintenance strategy, the goal is to develop an adaptive, intelligent fault diagnosis system that ensures early fault detection and enhances the reliability and efficiency of industrial operations.

8) Conclusion:

In conclusion, this project successfully demonstrates the development of an intelligent fault detection system for mechanical systems, with a primary focus on identifying bearing faults through vibration and Motor Current Signature Analysis (MCSA). By analyzing frequency-domain characteristics and current signature variations, we established a reliable foundation for diagnosing bearing defects, particularly in outer rings. Machine learning models were developed and evaluated to enhance the diagnostic process, with the Decision Tree model achieving the highest accuracy at 96.0%, followed by the Artificial Neural Network (ANN) at 94.0%, and the Support Vector Machine (SVM) at 88.0%. These results highlight the effectiveness of ML in fault classification and automation. Looking ahead, these models will be integrated into a real-time predictive maintenance framework to enable early fault detection, reduce unplanned downtimes, and improve overall system reliability. This project lays a strong foundation for the implementation of smarter, data-driven maintenance strategies in industrial environments.

9) References

1.) Detection of Bearing Faults in Mechanical Systems Using Stator Current Monitoring- Sukhjeet Singh and Navin Kumar

2.) Motor Current Signature Analysis for Bearing Fault Detection in Mechanical Systems- Sukhjeet Singh , Amit Kumar, Navin Kumar

3.)https://ncd.io/blog/bearing-fault-detection-vibration-analysis/

4.)Immovilli F, Bellini A, Rubini R, et al. (2010) Diagnosis of Bearing Faults in Induction Machines by Vibration or Current Signals: A Critical Comparison. Industry Applications,